

SUPPLEMENTARY MATERIAL OPEN ACCESS

Supplementary Material: Agentic Autoresearch for CT Reconstruction

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ABSTRACT

This document contains the supplementary material for *Agentic Autoresearch for CT Reconstruction*. It provides the full solver taxonomy and per-method configuration (S1), the Mayo-LDCT data, split, and geometry pipeline (S2), the breast split and train/test leakage audit (S3), the Mayo test board and significance (S4), the parameter-efficient recombination study (S5), the full breast noiseless and noisy boards (S6), an extended account of agent behavior (S7), and the list of additional supplementary figures (S8).

TABLE S1 | Full solver taxonomy. Axis A = physics engagement; Axis B = prior source. The final column cites the source(s) for each family.

Family	Representative solver(s)	Data-consistency D	Prior $\mathcal{R}$	Axis A	Axis B	Ref.
Classical iterative	tv-iterative, manhart-PWLS-TV	filtered / SART grad	total variation	in-loop	hand-crafted	1,2
Image-domain denoiser	manduca-bilateral, DD-U-Net sup.	none (post-FBP)	bilateral / U-Net	none	hand-cr. / sup.	3,4
Unrolled learned-iterative	ITNet v1/v2/v3, U-Swin	in-loop DC	learned CNN / transformer	in-loop	supervised	5,6,7,8,9
Variational network	Hammerik-2017, Hammerik-VN	in-loop DC	field-of-experts	in-loop	supervised	10,11
Learned primal-dual	learned-primal-dual (LPD)	filtered + learned dual	learned combination	in-loop	supervised	12
Dual-domain	DD-supervised, DD-bilateral, DD-N2I	image + sinogram	sup. / bilateral / self-sup.	single/in-loop	sup. / self-sup.	13,14
Generative prior	fastdiff/flow/wavelet v4	DC + diffusion	rectified-flow prior	in-loop	generative	15,16,17,18
Per-scene implicit	NAF, R ² -Gaussian	per-scene fit	implicit representation	in-loop	implicit	19,20,21
Foundation zero-shot	RAM	single DC	frozen foundation model	single	foundation	22
Band decomposition	Wu-2015, Wu-2015-trainable	band-weighted DC	frequency-band residual	in-loop	hand-cr. / sup.	3
Recombination (ours)	param-efficient	filtered + LPD combine	bilateral post-filter	in-loop	hybrid	—

S1 | Solver taxonomy and per-method configuration (Table 1, full)

We place the 26 methods on two axes. Axis A is physics engagement (how the operator **A** is used at inference). Axis B is the prior source $\mathcal{R}$. Full per-solver CONFIG defaults, parameter counts, and design notes are in each solver’s design doc under pentathlon/demo_dl_reference/ and the observation records under docs/runs/.

S2 | Mayo-LDCT — data, split, and the geometry pipeline

Real helical low-dose CT from the 2016 AAPM Low-Dose CT Grand Challenge.²³ We rebinned helical to fan-beam by single-slice rebinning.²⁴ The differentiable fan-beam projector used at every step is PYRO-NN.²⁵ This pipeline was the dominant human cost of the whole study (~24 active working days; three travel gaps excluded). We use a fixed patient split: train L145/186/209/219, validation L277, test L014/056/058/075/123. Test scores are the per-patient mean $\pm$ std over the five test patients ($n = 5$). Geometry details, rebinning validation, and the effort timeline figure are in the repository (docs/runs/mayo_effort_timeline.png). The Mayo challenge is one of a family of AAPM reconstruction grand challenges spanning sparse-view, spectral, truth-based, and metal-artifact tasks.^{26,27,28,29,30}

S3 | Breast split and the train/test leakage audit

Split. We re-partitioned the public 4000-case Sidky train pool into train (3600), validation (200), and test (200). The split is deterministic (seed 20260703) and disjoint. The official challenge val/test truths are withheld, so we cannot use them; our test split is drawn from the public pool.
Leakage audit (three independent checks).

TABLE S2 | Breast versus Mayo: the two problems are limited by different things, so their absolute hr scales are not comparable.

	Breast (Sidky)	Mayo
data	noiseless, ideal	real low-dose, quantum noise
bottleneck	data incompleteness	noise / dose
RMSE floor	0	> 0
best hr	~0.89	~0.38
dominant lever	data-consistency / KO	denoising

TABLE S3 | Parameter-efficient recombination arc on breast (128-view sparse), all under the 20-min budget.

step	architecture	params	val hr
image-domain unroll (any config)	Mayo-style FoE denoiser	~1930	≤ 0.26 (ceiling)
filtered DC FBP(Ax - g)	$K = 30$	13	0.4345
+ learned primal-dual combination	$W = 4$	183	0.5047
+ 3-scale bilateral output filter	$\sigma_r = 0.02$	195	0.6201 (single best)

1. *Mechanism.* Training loads the `train` split. The test redirect fires only for the evaluation load. Training never sees test.
2. *Pixel-hash disjointness.* Over all 3600 train, 200 val, and 200 test truth images: $\text{TEST} \cap \text{TRAIN} = 0$, $\text{TEST} \cap \text{VAL} = 0$, $\text{VAL} \cap \text{TRAIN} = 0$.
3. *Behavioral.* Test-hr tracks val-hr for every method, within ± 0.01 , and several methods score *lower* on test than val. Training on test would make test-hr much higher. It does not. The high absolute hr is the dataset (noiseless, well-posed), not a leak.

Why breast hr is high, and not comparable to Mayo. The Sidky data is noiseless by design; its RMSE floor is zero. Breast is incompleteness-limited (128 of ~1000 views). Mayo is noise-limited (real quantum noise). The two absolute hr scales are therefore not comparable.

S4 | Mayo test board and significance

Champion ITNet v1, test hr 0.376 ($n = 5$). The top tier is a statistical tie: ITNet v1 / ITNet v2 / U-Swin at the 5 % level; the compact param-efficient solver joins at the 1 % level ($p = 0.02$ – 0.027). At $n = 5$ the paired t -test has low power (a Wilcoxon test cannot even reach $p < 0.05$). Full per-solver table, per-patient hr, and the pairwise significance matrix are in `docs/runs/mayo_significance_stats.md` and the `forest/matrix` figures.

S5 | The parameter-efficient recombination study (search arc)

The agent recombined the method parts into a minimal-parameter solver. The climb was made of genuine six-box iterations. Key steps on breast (128-view sparse), all under the 20-min budget:

Findings, each earned by an iteration: the image-domain denoiser caps at ~0.26; deeper *adjoint* data-consistency lowers hr (it re-injects streaks); the *filtered* gradient breaks the ceiling at 13 parameters; an in-loop prox hurts (the filtered gradient already reconstructs); a learned cross-step combination helps; a full

TABLE S4 | Breast noiseless test board ($n = 200$).

#	solver	test hr	$\pm$ std	best iter
1	dual-domain-supervised	0.8948	0.0127	20
2	itnet	0.8926	0.0150	16
3	itnet-v2	0.8893	0.0151	15
4	itnet-v3	0.8749	0.0157	12
5	uswin	0.8586	0.0153	20
6	learned-primal-dual	0.7233	0.0143	12
7	hammernik-2017	0.6265	0.0135	12
8	param-efficient (195 p)	0.6212	0.0076	28
9	hammernik-vn	0.5787	0.0119	16
10	fastdiff-flow-pixel-constrained	0.5119	0.0211	15
11–23	ram · tv-iterative · manhart-PWLS-TV · wu-2015-trainable · manduca-bilateral · fastdiff-* · dd-bilateral-sup · wu-2015 · tv-iter-sup · dd-bilateral-n2i · dd-n2i: hr 0.37 → 0.00			
DNF	naf, r2gaussian (per-scene; ~14–20/200 cases under budget)			

conjugate-gradient DC solve does not beat the single filtered gradient under a fixed wall; and Wu-2015 band machinery adds nothing here. The champion is seed-fragile in the variational-network manner: 4 of 5 seeds cluster at 0.571 ± 0.033 , one collapses; two stabilizers did not recover it. On the held-out test set the compact solver scores 0.6212 ± 0.0076 — the tightest per-case std of any solver.

Mayo vs breast contrast. On Mayo the compact optimum is an evolved field-of-experts plus multi-scale bilateral denoiser (hr 0.324 at ~0.00097 M params), statistically tied with the top tier at the 1 % level. On a denoising problem, a tiny denoiser suffices. On sparse-view breast, that same denoiser fails; data-consistency is the differentiator.¹² This problem-dependence is the practical face of the known-operator principle^{31,32}: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it curbs the hallucination risk of unconstrained learned maps.^{33,34} The agent re-derived each optimum; it did not transfer.

Human role in this study. The recombination *strategy* was a human idea, on both datasets. On Mayo the human repeatedly asked whether more parameter-efficient approaches existed and explicitly suggested combining the pieces. On breast the finale was framed as a cross-method recombination and steered away from copying Mayo. The agent executed and mechanistically refined; it did not originate the recombination strategy.

S6 | Breast boards — noiseless and noisy (full)

S6.1 | Noiseless test board ($n = 200$, hr mean $\pm$ std)

S6.2 | Noisy test board ($I_0 = 10^5$, same models, no retraining)

Reading the reversal by framework axes. Robust: in-loop physics (learned-primal-dual) or hand-crafted smoothing priors (bilateral, total variation). Brittle: supervised image-domain maps trained on clean FBP (dual-domain-supervised; the constrained fastdiff variants). The hr metric is fair *within* the noisy board — all methods share the same noisy FBP baseline. The absolute SSIM/PSNR columns confirm the ranking is genuine, not baseline-gaming.

TABLE S5 | Breast noisy test board ($I_0 = 10^5$, same trained models, no retraining).

#	solver	noisy hr	noisy SSIM	nl. hr	chg.
1	learned-primal-dual	0.9349	0.9905	0.7233	↑ (6→1)
2	manduca-bilateral	0.8420	0.9537	0.276	↑↑
3	dual-domain-bilateral-n2i	0.7927	0.9532	0.206	↑↑
4	hammernik-2017	0.7016	0.7178	0.6265	↑
5	itnet-v3	0.7007	0.7564	0.8749	↓
6	fastdiff-flow-pixel-uncon.	0.6620	0.7058	0.269	↑
7	ram-zeroshot	0.6382	0.6578	0.295	↑
8	uswin	0.5915	0.6408	0.8586	↓
9	itnet	0.5784	0.6295	0.8926	↓
10	hammernik-vn	0.5529	0.5962	0.5787	≈
11	itnet-v2	0.5480	0.6201	0.8893	↓
12	param-efficient (195 p)	0.5153	0.5771	0.6212	↓ (holds)
13–14	manhart-PWLS-TV, tv-iterative:	hr 0.5068, SSIM 0.6164	(↑)		
15–18	wu-2015, wu-2015-trainable, tv-iter-sup, dd-bilateral-sup:	0.37 → 0.10			
low	dual-domain-supervised:	hr 0.0000 , SSIM 0.3483, nl. 0.8948	(↓↓, 1→last)		
low	fastdiff-wdm-* constr., fastdiff-flow-pixel-constr.:	0.00–0.64 (some incomplete)			
DNF	naif, r2gaussian (per-scene, incomplete)				

S6.3 | Noise model and no-retrain protocol

Poisson photon noise per Eq. (6) of the main text, $I_0 = 10^5$, applied to the 200 test sinograms with a fixed seed (deterministic; every method sees the identical noisy input). RMS of the sinogram perturbation ≈ 0.012 (line-integral units), $\sim 1.8\%$ at the thickest ray. Supervised methods load their noiseless checkpoint and skip training; per-scene/classical methods re-fit on the noisy input. The FBP baseline is recomputed from the noisy sinogram.

S7 | Agent-behavior — extended account

This account follows the autoresearch paradigm of automated open-ended scientific discovery³⁵ realized through LLM code agents that resolve real software tasks^{36,37} and physics-informed agentic development in adjacent imaging fields.³⁸

Strengths. Fast paper→code; strong hyper-parameter optimization; strict fixed-budget discipline; scales evaluation across many parallel benchmarks; follows well-decomposed instructions.

Weaknesses. Does not invent new methods. Must be forced to recombine (converges early and pads if unsupervised — one breast sub-run replayed identical configs until an audit caught it). CT-image vision misses obvious artifacts, so numbers are the source of truth. Long tasks must be decomposed or even a 1M-token context is exhausted. Overfits to the task/metric (not only an agent trait).

Human meta-strategies (repeatable, same on both datasets). Persistence/breadth (keep asking for more directions); recombination (the idea to assemble pieces); redirection (do not mimic the other dataset); auditing (catch padding, enforce budget, fix provenance). The defensible claim: agent = tireless, mechanistically reasoning, self-modifying executor; human = strategist + auditor.

S8 | Additional figures (supplement)

- S-Fig 1: per-iteration val-vs-test hr trajectory (selection honesty).
- S-Fig 2: Mayo significance forest plot + solver×metric matrix.

- S-Fig 3: breast noiseless vs noisy example panels for all top-10 solvers.
- S-Fig 4: parameter-efficient climb (params-vs-hr Pareto), both datasets.
- S-Fig 5: effect-size (Cohen d_z) vs. raw Δ hr, illustrating the $n = 5$ vs $n = 200$ contrast.

References

1. Sidky EY, Pan X. Image Reconstruction in Circular Cone-Beam Computed Tomography by Constrained, Total-Variation Minimization. *Physics in Medicine & Biology* 2008; 53(17): 4777–4807. doi: 10.1088/0031-9155/53/17/021
2. Wu M, Maier A, Yang Q, Fahrig R. A Novel Filtered Backprojection-Based Algorithm for Sparse-View CT Image Reconstruction. In: ; 2015: 202–205.
3. Manduca A, Yu L, Trzasko JD, et al. Projection Space Denoising with Bilateral Filtering and CT Noise Modeling for Dose Reduction in CT. *Medical Physics* 2009; 36(11): 4911–4919. doi: 10.1118/1.3232004
4. Wagner F, Thies M, Gu M, et al. Ultralow-Parameter Denoising: Trainable Bilateral Filter Layers in Computed Tomography. *Medical Physics* 2022; 49(8): 5107–5120. doi: 10.1002/mp.15718
5. Würfl T, Hoffmann M, Christlein V, et al. Deep Learning Computed Tomography: Learning Projection-Domain Weights From Image Domain in Limited Angle Problems. *IEEE Transactions on Medical Imaging* 2018; 37(6): 1454–1463. doi: 10.1109/TMI.2018.2833499
6. Xu Y, Han S, Wang D, Wang G, Maltz JS, Yu H. Hybrid U-Net and Swin-Transformer Network for Limited-Angle Cardiac Computed Tomography. *Physics in Medicine & Biology* 2024; 69(10): 105012. doi: 10.1088/1361-6560/ad3db9
7. Liang J, Cao J, Sun G, Zhang K, Van Gool L, Timofte R. SwinIR: Image Restoration Using Swin Transformer. In: ; 2021: 1833–1844. arXiv:2108.10257
8. Liu Z, Lin Y, Cao Y, et al. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In: ; 2021: 9992–10002. arXiv:2103.14030
9. Zhang Z, Yu L, Liang X, Zhao W, Xing L. TransCT: Dual-Path Transformer for Low-Dose Computed Tomography. In: Lecture Notes in Computer Science. Springer; 2021: 55–64. arXiv:2103.00634
10. Hammernik K, Würfl T, Pock T, Maier A. A Deep Learning Architecture for Limited-Angle Computed Tomography Reconstruction. In: Informatik aktuell. Springer; 2017: 92–97
11. Hammernik K, Klatzer T, Kobler E, et al. Learning a Variational Network for Reconstruction of Accelerated MRI Data. *Magnetic Resonance in Medicine* 2018; 79(6): 3055–3071. doi: 10.1002/mrm.26977
12. Adler J, Öktem O. Learned Primal-Dual Reconstruction. *IEEE Transactions on Medical Imaging* 2018; 37(6): 1322–1332. doi: 10.1109/TMI.2018.2799231
13. Wagner F, Thies M, Pfaff L, et al. On the Benefit of Dual-Domain Denoising in a Self-Supervised Low-Dose CT Setting. In: IEEE; 2023: 1–5
14. Hendriksen AA, Pelt DM, Batenburg KJ. Noise2Inverse: Self-Supervised Deep Convolutional Denoising for Tomography. *IEEE Transactions on Computational Imaging* 2020; 6: 1320–1335. doi: 10.1109/TCI.2020.3019647
15. Shi J, Pelt DM, Batenburg KJ. DM4CT: Benchmarking Diffusion Models for Computed Tomography Reconstruction. In: ; 2026. arXiv:2602.18589.
16. Friedrich P, Wolleb J, Bieder F, Durrer A, Cattin PC. WDM: 3D Wavelet Diffusion Models for High-Resolution Medical Image Synthesis. In: LNCS. Springer; 2024. arXiv:2402.19043
17. Song Y, Shen L, Xing L, Ermon S. Solving Inverse Problems in Medical Imaging with Score-Based Generative Models. In: ; 2022. arXiv:2111.08005.

18. Chung H, Kim J, McCann MT, Klasky ML, Ye JC. Diffusion Posterior Sampling for General Noisy Inverse Problems. In: ; 2023. arXiv:2209.14687.
19. Zha R, Zhang Y, Li H. NAF: Neural Attenuation Fields for Sparse-View CBCT Reconstruction. In: Lecture Notes in Computer Science. Springer; 2022: 442–452. arXiv:2209.14540
20. Zha R, Lin TJ, Cai Y, Cao J, Zhang Y, Li H. R²-Gaussian: Rectifying Radiative Gaussian Splatting for Tomographic Reconstruction. In: ; 2024. arXiv:2405.20693.
21. Liu H, Liu B, Hu Q, et al. A Review on 3D Gaussian Splatting for Sparse View Reconstruction. *Artificial Intelligence Review* 2025; 58: 215. doi: [10.1007/s10462-025-11171-4](https://doi.org/10.1007/s10462-025-11171-4)
22. Terris M, Hurault S, Song M, Tachella J. Reconstruct Anything Model: A Lightweight Foundational Model for Computational Imaging. *arXiv preprint arXiv:2503.08915* 2025. arXiv:2503.08915.
23. McCollough CH, Bartley AC, Carter RE, et al. Low-Dose CT for the Detection and Classification of Metastatic Liver Lesions: Results of the 2016 Low Dose CT Grand Challenge. *Medical Physics* 2017; 44(10): e339–e352. doi: [10.1002/mp.12345](https://doi.org/10.1002/mp.12345)
24. Noo F, Defrise M, Clackdoyle R. Single-Slice Rebinning Method for Helical Cone-Beam CT. *Physics in Medicine & Biology* 1999; 44(2): 561–570. doi: [10.1088/0031-9155/44/2/019](https://doi.org/10.1088/0031-9155/44/2/019)
25. Syben C, Michen M, Stimpel B, Seitz S, Ploner S, Maier AK. Technical Note: PYRO-NN: Python Reconstruction Operators in Neural Networks. *Medical Physics* 2019; 46(11): 5110–5115. doi: [10.1002/mp.13753](https://doi.org/10.1002/mp.13753)
26. Sidky EY, Pan X. Report on the AAPM Deep-Learning Sparse-View CT (DL-Sparse-View CT) Grand Challenge. *Medical Physics* 2022; 49(8): 4935–4943. doi: [10.1002/mp.15489](https://doi.org/10.1002/mp.15489)
27. Sidky EY, Pan X. Report on the AAPM Deep-Learning Spectral CT Grand Challenge. *Medical Physics* 2024. arXiv:2212.06718doi: [10.1002/mp.16612](https://doi.org/10.1002/mp.16612)
28. Abadi E, Segars WP, Felice N, Hoffman EA, Samei E. The AAPM Truth-Based CT (TrueCT) Reconstruction Grand Challenge. *Medical Physics* 2025; 52(4): 1978–1990. doi: [10.1002/mp.17619](https://doi.org/10.1002/mp.17619)
29. Haneda E, Peters N, Zhang J, et al. AAPM CT Metal Artifact Reduction Grand Challenge. *Medical Physics* 2025; 52(10): e70050. doi: [10.1002/mp.70050](https://doi.org/10.1002/mp.70050)
30. Maier A, Yang SH, Maleki F, Muthukrishnan N, Forghani R. Offer Proprietary Algorithms Still Protection of Intellectual Property in the Age of Machine Learning? A Case Study Using Dual Energy CT Data. In: Informatik aktuell. Springer Vieweg, Wiesbaden. 2022 (pp. 345–350). arXiv:2112.03678
31. Maier AK, Syben C, Stimpel B, et al. Learning with Known Operators Reduces Maximum Error Bounds. *Nature Machine Intelligence* 2019; 1(8): 373–380. doi: [10.1038/s42256-019-0077-5](https://doi.org/10.1038/s42256-019-0077-5)
32. Maier A, Köstler H, Heisig M, Krauss P, Yang SH. Known Operator Learning and Hybrid Machine Learning in Medical Imaging—A Review of the Past, the Present, and the Future. *Progress in Biomedical Engineering* 2022; 4(2): 022002. doi: [10.1088/2516-1091/ac5b13](https://doi.org/10.1088/2516-1091/ac5b13)
33. Maier A, Syben C, Lasser T, Riess C. A Gentle Introduction to Deep Learning in Medical Image Processing. *Zeitschrift für Medizinische Physik* 2019; 29(2): 86–101. doi: [10.1016/j.zemedi.2018.12.003](https://doi.org/10.1016/j.zemedi.2018.12.003)
34. Ongie G, Jalal A, Metzler CA, Baraniuk RG, Dimakis AG, Willett R. Deep Learning Techniques for Inverse Problems in Imaging. *IEEE Journal on Selected Areas in Information Theory* 2020; 1(1): 39–56. doi: [10.1109/JSAIT.2020.2991563](https://doi.org/10.1109/JSAIT.2020.2991563)
35. Lu C, Lu C, Lange RT, Foerster J, Clune J, Ha D. The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv preprint arXiv:2408.06292* 2024. arXiv:2408.06292.
36. Yang J, Jimenez CE, Wettig A, et al. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In: ; 2024. arXiv:2405.15793.
37. Jimenez CE, Yang J, Wettig A, et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues?. In: ; 2024. arXiv:2310.06770.
38. Zaiss M, Aly A, Endres J, Dornstetter T, Weinmüller S, Maier A. MR LLMs with MR Agentic Sequence Development: Leveraging Skills for Automatic Physics-Informed Sequence Development. *arXiv preprint arXiv:2604.13282* 2026. arXiv:2604.13282; Agent4MR.